Mode 6 — Baseline-BFP

DP32 (8×8) → DP16 (8×8) → DP16 (8×4) → DP8 (8×4) → block exponents

Step 1 — DP32 (8×8)

$$\mathrm{DP32}\,(8{\times}8) = \sum_{i=0}^{31} a_i\, b_i$$

Step 2 — DP16 (8×8)

$$\mathrm{DP32}\,(8{\times}8) = \underbrace{\sum_{i=0}^{15} a_i\, b_i}_{\mathrm{DP16}\,(8\times 8)} + \underbrace{\sum_{i=16}^{31} a_i\, b_i}_{\mathrm{DP16}\,(8\times 8)}$$

Step 3 — DP16 (8×4)

$$\mathrm{DP32}\,(8{\times}8) = 2^4 \underbrace{\sum_{i=0}^{15} a_i\, b_i^{hi}}_{\mathrm{DP16}\,(8\times 4)} + \underbrace{\sum_{i=0}^{15} a_i\, b_i^{lo}}_{\mathrm{DP16}\,(8\times 4)} + 2^4 \underbrace{\sum_{i=16}^{31} a_i\, b_i^{hi}}_{\mathrm{DP16}\,(8\times 4)} + \underbrace{\sum_{i=16}^{31} a_i\, b_i^{lo}}_{\mathrm{DP16}\,(8\times 4)}$$

Step 4 — DP8 (8×4)

$$\begin{aligned}\mathrm{DP32}\,(8{\times}8) = {} & 2^4 \underbrace{\sum_{i=0}^{7} a_i\, b_i^{hi}}_{\mathrm{DP8}\,(8\times 4)} + 2^4 \underbrace{\sum_{i=8}^{15} a_i\, b_i^{hi}}_{\mathrm{DP8}\,(8\times 4)} + \underbrace{\sum_{i=0}^{7} a_i\, b_i^{lo}}_{\mathrm{DP8}\,(8\times 4)} + \underbrace{\sum_{i=8}^{15} a_i\, b_i^{lo}}_{\mathrm{DP8}\,(8\times 4)} \\ & + 2^4 \underbrace{\sum_{i=16}^{23} a_i\, b_i^{hi}}_{\mathrm{DP8}\,(8\times 4)} + 2^4 \underbrace{\sum_{i=24}^{31} a_i\, b_i^{hi}}_{\mathrm{DP8}\,(8\times 4)} + \underbrace{\sum_{i=16}^{23} a_i\, b_i^{lo}}_{\mathrm{DP8}\,(8\times 4)} + \underbrace{\sum_{i=24}^{31} a_i\, b_i^{lo}}_{\mathrm{DP8}\,(8\times 4)}\end{aligned}$$

Step 5 — Block exponents (one exponent block per DP8 (8×4))

$$\Big(2^4 \sum_{i=8q}^{8q+7} a_i b_i^{hi} + \sum_{i=8q}^{8q+7} a_i b_i^{lo},\ E_q\Big)_{q=0..3} \longrightarrow \Big(M,\ E\Big) \qquad E_q = E_{a,q} + E_{b,q}$$

$E = \max(E_0, \dots, E_3)$, $\delta_q = E - E_q$ (staged in the tree, the stage shifts telescope); truncating shifts, on block sums only; $\mathrm{DP32}\,(8{\times}8) = M \cdot 2^E$.

Step 6 — Result

$$\boxed{\begin{aligned} M = {} & 2^{-\delta_0}\Big(2^4 \sum_{i=0}^{7} a_i b_i^{hi} + \sum_{i=0}^{7} a_i b_i^{lo}\Big) + 2^{-\delta_1}\Big(2^4 \sum_{i=8}^{15} a_i b_i^{hi} + \sum_{i=8}^{15} a_i b_i^{lo}\Big) \\ & + 2^{-\delta_2}\Big(2^4 \sum_{i=16}^{23} a_i b_i^{hi} + \sum_{i=16}^{23} a_i b_i^{lo}\Big) + 2^{-\delta_3}\Big(2^4 \sum_{i=24}^{31} a_i b_i^{hi} + \sum_{i=24}^{31} a_i b_i^{lo}\Big), \\ E = {} & \max(E_0, E_1, E_2, E_3) \end{aligned}}$$